Instructions

To successfully complete this assessment, make sure to follow these instructions:

- Carefully read and understand each question before answering it.
- Respond to ALL questions in the assessment.
- Explicitly display your work and outline the steps taken for each answer. Support your explanations and responses with examples to showcase a comprehensive understanding. Partial marks will be granted for accurate work, even if the final answer is incorrect.
- Take note of the indicated weighting for each assessment attribute, denoted in 'Marking Rubric' at the end.

Submission

- Submit your completed assessment online using the following guidelines:
- Name and save your files as "ICC104_Report1_LastName_FirstName"
- Utilize the Assessment Submission link available under the Assessment tab in the primary navigation menu.
- Access your Learning Facilitator's feedback through the 'Marks' in the Learning Management System (LMS) portal.

Learning Outcomes

This assignment covers the following learning outcomes in component ICC104:

a. Explain the concept of Cloud and the traditional deployment model for IT services.
b. Compare and explain essential characteristics of Cloud.
c. Outline and differentiate different service models of Cloud.
d. Explain and apply knowledge of various cloud deployment models and differentiate between them.

Assignment Overview

You are required **to write a 1500-word report based on the four provided case studies**. Each case study will focus on a different business, and the report will evaluate cloud adoption strategies, challenges, and the outcomes for each case.

Report Format

Your report should be structured as follows:

Introduction (250 words max.)

Provide an overview of cloud computing, emphasizing its role in transforming businesses.

Main Body (1000 words max.)

Address the following areas/questions of each case study:

- Briefly describe the organization, its industry, and the role cloud computing plays in its operations.
- What specific operational challenges or limitations did the organization face prior to cloud adoption?
- How did the organization solve these challenges using cloud technology? Highlight the key services, tools, and cloud platforms utilized.
- Identify and explain the cloud service models (IaaS, PaaS, SaaS) used in the case and their role in the solution.
- Discuss the deployment models employed (public, private, hybrid) and how they suited the organization's requirements.
- What were the benefits, and were there any areas for improvement?

Conclusion (250 words max.)
Summarize the key insights from the case studies and reflect on how cloud computing is transforming industries. Discuss any common trends observed and suggest areas for further investigation or technological improvement in cloud adoption.

Case Study Summaries:

Case Study 1: Netflix Enhances Global Streaming with AWS
Netflix, a global leader in video streaming, relies heavily on the cloud to deliver content to millions of users worldwide. Facing challenges related to scalability and the demand for high availability, Netflix adopted Amazon Web Services (AWS) to enhance its infrastructure and ensure a seamless streaming experience.
Read more: https://aws.amazon.com/solutions/case-studies/netflix/

Case Study 2: Tiger Brokers collaborates with Alibaba Cloud to enhance its operations and services.
Tiger Brokers, an online brokerage firm, used Alibaba Cloud to enhance its technology infrastructure. By leveraging cloud computing, Tiger Brokers optimized its data processing capabilities, enabling the firm to provide global investors with real-time trading services. The Alibaba Cloud solutions helped Tiger Brokers improve operational efficiency, scale their services quickly, and ensure security and stability for users, particularly during peak trading times. The adoption of these cloud services enabled the company to enhance its overall user experience and service reliability.
Read more: https://www.alibabacloud.com/en/customers/tiger-brokers

Case Study 3: New Zealand moves to a cloud-based identity solution
The New Zealand Department of Internal Affairs (DIA) adopted Microsoft Azure Active Directory (Azure AD) to modernize its identity management system and streamline secure access to government services. This moves enabled DIA to offer enhanced cybersecurity, improved user experience, and easier collaboration across agencies.
Read more: https://customers.microsoft.com/en-us/story/1436689624484208913-new-zealand-dia-government-azure-active-directory

Case Study 4: Etihad Airways focused its business challenges and worked with IBM
Etihad Airways partnered with IBM to enhance its digital transformation by leveraging IBM Cloud services. The airline implemented IBM's hybrid cloud solutions to optimize operations, improve customer experiences, and increase agility.
Read more: https://www.ibm.com/case-studies/etihad-airways-ibm-cloud

Citing and Referencing:

Use proper APA style for citing and referencing research is imperative. For more details on referencing guidelines, please refer to the MDS Academic Skills webpage (https://mediadesignschool.libguides.com/apa/tool). Be sure to cite any lecture notes utilized, along with at least five additional sources pertinent to the subject area, based on sources and further research.

Academic Integrity:

You are required to familiarize yourself with the MDS Academic Integrity Policy and Procedure (https://mediadesignschool.libguides.com/academic_integrity), including the consequences for academic misconduct, which are accessible online. Students must also retain copies of all submitted materials and any assessment drafts.